Data quality and contracts

Data engineering · Level 4 · Lesson 3



In this lesson you will

- Write quality tests as code: types, nulls, ranges, uniqueness, references, freshness.
- Agree a data contract between producers and users.
- Decide what a failed test does.

Key ideas

- A test is a query that should return nothing.
- The contract is the agreement, as code.
- Block, quarantine or alert, by cost.

Contract and tests

1. Choose (3 min): a dataset
2. Contract (6 min): write the rules
3. Test (5 min): rules become tests
4. Decide (3 min): block, quarantine or alert?

Swap contracts with a partner and try to write a row that breaks theirs but passes every test.

Let’s discuss

1. Why use a LEFT JOIN to find orphans?
2. When is quarantine better than blocking the load?
3. What can’t any test catch?

Exit ticket

- Write one rule of a data contract and the SQL test that checks it.